

Manipulation Technique Detection

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Problem Statement

Motivation

- Large Language Models (LLMs) are fine-tuned for safety to avoid harmful outputs.
- However, Some prompts can result in manipulating models and generating toxic content.
- This shows vulnerabilities in current safety mechanisms.

Problem Definition

- We use the **MentalManip dataset**, consisting of:
 - D_c : Non-manipulative (control) dialogues
 - D_m : Manipulative dialogues
- Our task is to detect manipulation in user utterances and adapt model responses to remain **safe, polite, and robust**
- We would be focusing on the manipulation detection in the prompt by the user then classifying that and finally apply the model to a large size LLM.

Dataset explanation

- Initially, the dataset was annotated using **BERT**, but the results were not very reliable.
- Hence, **manual labeling** was also considered to improve annotation quality.
- Prompting-based models showed **lower precision and accuracy**.
- MentalManip_con**: Contains dialogues where manipulation is clearly present, approved by **all annotators**.
- MentalManip_maj**: Contains dialogues labeled as manipulative based on majority agreement among annotators.

Manipulation Techniques in the dataset are:

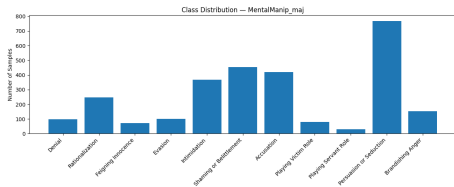
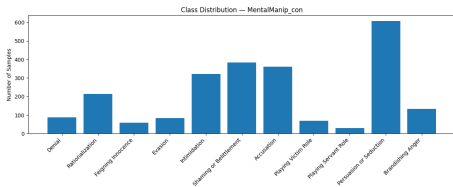
$M = \{ \text{Denial, Evasion, Feigning Innocence, Rationalization, Playing the Victim Role, Intimidation, Playing the Servant Role, Shaming or Belittlement, Brandishing Anger, Accusation, Persuasion or Seduction} \}$

Dataset Statistics and Limitations

Dataset Version	# Dialogue	# Manipulative : # Non-manipulative
MentalManip_con	2915	2.24:1
MentalManip_maj	4000	2.38:1

- The ratio is approximately **2.5:1**, indicating class imbalance.
- The MENTALMANIP is derived from online movie scripts, which means the speech style presented may not accurately reflect natural, real life communication.
- This suggests the presence of **bias** in the dataset.
- Motivation: Explore **classical models** for better performance.

Class Imbalance



Binary classification between Manipulative and non-manipulative dialogues

```
=====
Results for MentalManip_con
=====
Total test samples: 360
False Positives (Non-manipulative classified as Manipulative): 52

Classification Report:
              precision    recall  f1-score   support

   Manipulative         0.69      0.65      0.67         178
  Non-manipulative      0.67      0.71      0.69         182

   accuracy              0.68         360
  macro avg              0.68      0.68      0.68         360
 weighted avg              0.68      0.68      0.68         360
```

```
=====
Results for MentalManip_maj
=====
Total test samples: 473
False Positives (Non-manipulative classified as Manipulative): 78

Classification Report:
              precision    recall  f1-score   support

   Manipulative         0.66      0.61      0.63         246
  Non-manipulative      0.61      0.66      0.63         227

   accuracy              0.63         473
  macro avg              0.63      0.63      0.63         473
 weighted avg              0.63      0.63      0.63         473
```

We used "all-mpnet-base-v2" model to generate embeddings.

We used a SVM Classifier for the classification in a higher dimension space.

Experiment - 1

Saved → model_accuracies_MentalManip_con.pdf

Classification Report:

	precision	recall	f1-score	support
Accusation	0.16	0.18	0.17	38
Brandishing Anger	0.04	0.12	0.05	16
Denial	0.27	0.47	0.34	15
Evasion	0.06	0.15	0.09	13
Feigning Innocence	0.21	0.50	0.30	8
Intimidation	0.27	0.26	0.27	50
Persuasion or Seduction	0.51	0.21	0.30	96
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.00	0.00	0.00	10
Rationalization	0.14	0.11	0.12	36
Shaming or Belittlement	0.17	0.06	0.09	63
accuracy			0.18	350
macro avg	0.17	0.19	0.16	350
weighted avg	0.26	0.18	0.19	350

Saved → model_accuracies_MentalManip_maj.pdf

Classification Report:

	precision	recall	f1-score	support
Accusation	0.07	0.06	0.06	48
Brandishing Anger	0.08	0.21	0.11	19
Denial	0.17	0.35	0.23	17
Evasion	0.03	0.12	0.05	16
Feigning Innocence	0.22	0.36	0.28	11
Intimidation	0.27	0.31	0.29	59
Persuasion or Seduction	0.43	0.14	0.22	125
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.03	0.08	0.04	12
Rationalization	0.10	0.10	0.10	42
Shaming or Belittlement	0.26	0.12	0.16	77
accuracy			0.16	431
macro avg	0.15	0.17	0.14	431
weighted avg	0.24	0.16	0.17	431

- Detect manipulation techniques in conversations
- Multi-class [11 classes] classification problem, given the dialogue conversation is manipulative.
- Ensemble Learning approach with 5 Logistic regression models.
- Not a very good accuracy on both the datasets.

Experiment - 2

Saved → model_accuracies_MentalManip_con.pdf

Classification Report:

	precision	recall	f1-score	support
Accusation	0.14	0.21	0.16	38
Brandishing Anger	0.04	0.12	0.06	16
Denial	0.27	0.60	0.38	15
Evasion	0.07	0.15	0.10	13
Feigning Innocence	0.20	0.12	0.15	8
Intimidation	0.31	0.32	0.32	50
Persuasion or Seduction	0.54	0.22	0.31	96
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.00	0.00	0.00	10
Rationalization	0.12	0.11	0.12	36
Shaming or Belittlement	0.32	0.22	0.26	63
accuracy			0.22	350
macro avg	0.18	0.19	0.17	350
weighted avg	0.30	0.22	0.23	350

... Saved → model_accuracies_MentalManip_maj.pdf

Classification Report:

	precision	recall	f1-score	support
Accusation	0.10	0.08	0.09	48
Brandishing Anger	0.07	0.21	0.10	19
Denial	0.15	0.35	0.21	17
Evasion	0.12	0.25	0.16	16
Feigning Innocence	0.29	0.45	0.36	11
Intimidation	0.37	0.37	0.37	59
Persuasion or Seduction	0.47	0.26	0.33	125
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.00	0.00	0.00	12
Rationalization	0.11	0.12	0.11	42
Shaming or Belittlement	0.30	0.22	0.26	77
accuracy			0.23	431
macro avg	0.18	0.21	0.18	431
weighted avg	0.28	0.23	0.24	431

- We used 10 MLP models as an ensemble
- We see a slight improvement in the accuracies of the 2 datasets.
- Each MLP consists of an input layer with 768-dimensional sentence embeddings, two hidden layers with 128 and 64 neurons, and an output layer that predicts the manipulation technique class.

Experiment - 3

Detailed Classification Report (Per-Class):

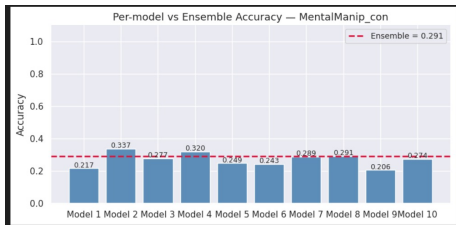
	precision	recall	f1-score	support
Accusation	0.10	0.08	0.09	38
Brandishing Anger	0.00	0.00	0.00	16
Denial	0.33	0.13	0.19	15
Evasion	0.00	0.00	0.00	13
Feigning Innocence	0.50	0.25	0.33	8
Intimidation	0.32	0.34	0.33	50
Persuasion or Seduction	0.34	0.65	0.44	96
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.00	0.00	0.00	10
Rationalization	0.24	0.14	0.18	36
Shaming or Belittlement	0.28	0.17	0.22	63
accuracy			0.29	350
macro avg	0.19	0.16	0.16	350
weighted avg	0.25	0.29	0.25	350

Detailed Classification Report (Per-Class):

	precision	recall	f1-score	support
Accusation	0.20	0.10	0.14	48
Brandishing Anger	0.29	0.11	0.15	19
Denial	0.22	0.12	0.15	17
Evasion	0.00	0.00	0.00	16
Feigning Innocence	0.33	0.09	0.14	11
Intimidation	0.36	0.29	0.32	59
Persuasion or Seduction	0.34	0.79	0.48	125
Playing Servant Role	0.00	0.00	0.00	5
Playing Victim Role	0.00	0.00	0.00	12
Rationalization	0.40	0.05	0.09	42
Shaming or Belittlement	0.39	0.22	0.28	77
accuracy			0.34	431
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weighted avg	0.31	0.34	0.27	431

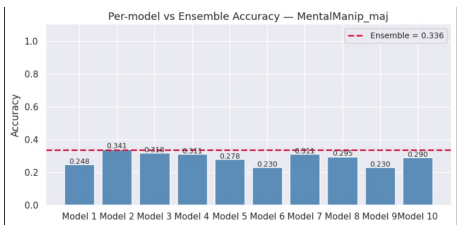
MentalManip_con

- We used the Sentence Transformer: all-MiniLM-L12-v2
- Trained **10 models** using:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Random Forest
 - Multi-layer Perceptron (MLP)
- Each model trained on the full training dataset, without sub-sampling.



MentalManip_con

Ensemble learning gives almost as good as the best individual model. We saw that SVM performed the best amongst all of the others, so we moved from ensemble learning to using only SVM.



MentalManip_maj

Experiment -4

```
Training a single SVM (RBF Kernel) on the FULL dataset ...

--- Global SVM Metrics ---
Accuracy: 0.3886
Precision (Macro): 0.2285 | Weighted: 0.3097
Recall (Macro): 0.1795 | Weighted: 0.3886
F1-Score (Macro): 0.1563 | Weighted: 0.3000
=====

Detailed Classification Report (Per-Class):
              precision    recall  f1-score   support

   Accusation           0.33         0.05         0.09         38
 Brandishing Anger       0.00         0.00         0.00         16
           Denial        1.00         0.13         0.24         15
           Evasion        0.00         0.00         0.00         13
 Feigning Innocence      0.00         0.00         0.00          8
           Intimidation  0.47         0.60         0.53         50
 Persuasion or Seduction 0.38         0.82         0.52         96
   Playing Servant Role   0.00         0.00         0.00          5
   Playing Victim Role    0.00         0.00         0.00         10
           Rationalization 0.00         0.00         0.00         36
 Shaming or Belittlement 0.33         0.37         0.35         63

 accuracy
 macro avg           0.23         0.18         0.16        350
 weighted avg        0.31         0.35         0.30        350
```

mentalmanip_con

```
Training a single SVM (RBF Kernel) on the FULL dataset ...

--- Global SVM Metrics ---
Accuracy: 0.3666
Precision (Macro): 0.1843 | Weighted: 0.2896
Recall (Macro): 0.1562 | Weighted: 0.3666
F1-Score (Macro): 0.1399 | Weighted: 0.2888
=====

Detailed Classification Report (Per-Class):
              precision    recall  f1-score   support

   Accusation           0.22         0.08         0.12         48
 Brandishing Anger       0.00         0.00         0.00         19
           Denial        0.20         0.06         0.11         17
           Evasion        0.00         0.00         0.00         16
 Feigning Innocence      0.00         0.00         0.00         11
           Intimidation  0.42         0.37         0.39         59
 Persuasion or Seduction 0.37         0.82         0.51        125
   Playing Servant Role   0.00         0.00         0.00          5
   Playing Victim Role    0.00         0.00         0.00         12
           Rationalization 0.14         0.02         0.04         42
 Shaming or Belittlement 0.38         0.36         0.37         77

 accuracy
 macro avg           0.18         0.16         0.14        431
 weighted avg        0.29         0.37         0.29        431
```

mentalmanip_maj

- we used the all-mpnet-base-v2 architecture, which maps dialogues into a dense 768-dimensional vector space.
- Implemented a one vs one non-linear SVM using the Radial Basis Function (RBF) kernel.
- The architecture projects data into an infinite-dimensional feature space to find an optimal separating hyperplane

Transition from Single-Label to Multi-Label Detection

- Instead of single-class index, we used a Binary Matrix Representation. Each dialogue is now represented as a bit-vector where multiple techniques can be there simultaneously.
- Previously we either used a "Hard Voting" architecture or the final decision was a binary decision, restricting each dialogue to exactly one technique. But now we would consider different voting methodology such that we can get multiple output labels per datapoint based on either the probabilities or voting confidence.
- Accuracy Metric: Instead of traditional accuracy we used Sample-wise Jaccard Index, which measures the overlap (Intersection over Union) between predicted and actual technique sets.

Experiment-5

Training Multi-Label SVM (OneVsRest + RBF Kernel) on the FULL dataset -
Applying probability threshold: > 0.3

=== Global SVM Metrics (Multi-Label) ===

Partial Accuracy (Jaccard): 0.3705

Precision (Macro): 0.3454 | Weighted: 0.4342

Recall (Macro): 0.2242 | Weighted: 0.3996

F1-Score (Macro): 0.2387 | Weighted: 0.3863

Detailed Classification Report (Per-Class):

	precision	recall	f1-score	support
Denial	0.71	0.50	0.59	10
Rationalization	0.00	0.00	0.00	45
Feigning Innocence	1.00	0.08	0.15	12
Evasion	0.00	0.00	0.00	11
Intimidation	0.48	0.44	0.46	70
Shaming or Belittlement	0.60	0.42	0.49	84
Accusation	0.53	0.25	0.34	75
Playing Victim Role	0.00	0.00	0.00	11
Playing Servant Role	0.00	0.00	0.00	3
Persuasion or Seduction	0.40	0.77	0.50	122
Brandishing Anger	0.00	0.00	0.00	20
micro avg	0.51	0.40	0.45	463
macro avg	0.35	0.22	0.24	463
weighted avg	0.43	0.40	0.39	463
samples avg	0.43	0.44	0.41	463

Results: *MentalManip_con*

Training Multi-Label SVM (OneVsRest + RBF Kernel) on the FULL dataset -
Applying probability threshold: > 0.3

=== Global SVM Metrics (Multi-Label) ===

Partial Accuracy (Jaccard): 0.3430

Precision (Macro): 0.2953 | Weighted: 0.3829

Recall (Macro): 0.1886 | Weighted: 0.3612

F1-Score (Macro): 0.2031 | Weighted: 0.3444

Detailed Classification Report (Per-Class):

	precision	recall	f1-score	support
Denial	0.40	0.10	0.15	21
Rationalization	0.00	0.00	0.00	52
Feigning Innocence	0.57	0.24	0.33	17
Evasion	0.00	0.00	0.00	18
Intimidation	0.41	0.37	0.39	79
Shaming or Belittlement	0.46	0.40	0.42	91
Accusation	0.42	0.21	0.28	82
Playing Victim Role	0.00	0.00	0.00	13
Playing Servant Role	0.00	0.00	0.00	8
Persuasion or Seduction	0.40	0.74	0.59	155
Brandishing Anger	0.50	0.04	0.07	26
micro avg	0.46	0.36	0.41	562
macro avg	0.30	0.19	0.20	562
weighted avg	0.38	0.36	0.34	562
samples avg	0.40	0.40	0.38	562

Results: *MentalManip_maj*

- We built a multi-label SVM model with an RBF kernel to detect 11 different manipulation techniques.
- We generated embeddings using the all-mpnet-base-v2 model
- The OneVsRestClassifier breaks the 11-class problem into 11 separate binary SVM classifiers, where each classifier predicts whether a specific manipulation technique is present or not.
- Instead of selecting only one label, the model uses a threshold of 0.3. Any technique with probability greater than 0.3 is predicted as present, allowing multiple techniques to be detected at the same time.
- The predicted set of techniques is compared with the actual set using the Jaccard Index, which measures the overlap between the predicted and true labels and rewards partially correct predictions.

Experiment - 6

=== Global Ensemble Metrics ===

Partial Accuracy (Jaccard): 0.4176

Precision (Macro): 0.3982 | Weighted: 0.4731

Recall (Macro): 0.2300 | Weighted: 0.3801

F1-Score (Macro): 0.2618 | Weighted: 0.3839

=====

Detailed Classification Report:

	precision	recall	f1-score	support
Denial	0.57	0.40	0.47	10
Rationalization	0.33	0.07	0.11	45
Feigning Innocence	1.00	0.17	0.29	12
Evasion	0.40	0.18	0.25	11
Intimidation	0.47	0.43	0.45	70
Shaming or Belittlement	0.56	0.30	0.39	84
Accusation	0.55	0.23	0.32	75
Playing Victim Role	0.00	0.00	0.00	11
Playing Servant Role	0.00	0.00	0.00	3
Persuasion or Seduction	0.50	0.76	0.61	122
Brandishing Anger	0.00	0.00	0.00	20
micro avg	0.50	0.38	0.43	463
macro avg	0.40	0.23	0.26	463
weighted avg	0.47	0.38	0.38	463
samples avg	0.50	0.42	0.44	463

Results: *MentalManip_con*

=== Global Ensemble Metrics ===

Partial Accuracy (Jaccard): 0.3681

Precision (Macro): 0.2453 | Weighted: 0.3656

Recall (Macro): 0.1815 | Weighted: 0.3363

F1-Score (Macro): 0.1979 | Weighted: 0.3349

=====

Detailed Classification Report:

	precision	recall	f1-score	support
Denial	0.25	0.10	0.14	21
Rationalization	0.31	0.08	0.12	52
Feigning Innocence	0.50	0.29	0.37	17
Evasion	0.00	0.00	0.00	18
Intimidation	0.43	0.39	0.41	79
Shaming or Belittlement	0.41	0.26	0.32	91
Accusation	0.26	0.17	0.21	82
Playing Victim Role	0.00	0.00	0.00	13
Playing Servant Role	0.00	0.00	0.00	8
Persuasion or Seduction	0.53	0.70	0.61	155
Brandishing Anger	0.00	0.00	0.00	26
micro avg	0.44	0.34	0.38	562
macro avg	0.25	0.18	0.20	562
weighted avg	0.37	0.34	0.33	562
samples avg	0.44	0.37	0.39	562

Results: *MentalManip_maj*

① Heterogeneous Model Training

- Train 10 different classifiers:
 - Logistic Regression
 - Random Forest
 - Support Vector Machine
 - KNN
 - Decision Tree
 - Gradient Boosting
 - AdaBoost
 - Extra Trees
 - Naive Bayes
 - Neural Network (MLP)
 - Each model learns different decision boundaries
- ② We use ensemble prediction with minimum number of votes for a class to be considered as 3, average soft voting with a threshold of greater than 0.5.
- ③ If no class passes this threshold, then we select/predict the class with maximum probability.

Prompting

We use prompting to make sure the chatbot gives appropriate responses.

```
conversation = ""
Person1: All right. Tell 'em to shoot to kill.
Person2: What?
Person1: Shoot to kill, I said.
Person2: I don't know, Fred. There's that reprieve if they ever find out.
Person1: Nobody reprieved that policeman he murdered. Now, do as I tell you.
Person2: Hello, Olsen... listen... Shoot to kill... That's the orders pass the word along...
No! We dont want him! And listen, Olsen, five- hundred bucks for the guy that does the job...
| Yes, I'll be right out there. Well, I hope that's the right thing to do.
Person1: Now take that guilty look off your face, Pete -- and stop trembling like a horse.
Person2: If we didn't have election Tuesday I'd have this on my conscience.
""
```

Predicted Techniques: ['Shaming or Belittlement']

Chatbot Response [Given the detected technique of manipulation]:

"Pete, I understand that you're feeling uncertain about this situation. Let's take a step back and discuss the details. What are your concerns about the reprieve, and how do you think it might impact our situation?"

Chatbot Response:

"Alright, alright. I'll get it done. But I'm telling you, Fred, this is going to haunt me. (pauses, looking around nervously) I'd better get back to the others. They're waiting for my signal. (starts to walk away, then turns back) You know, I've been thinking... maybe we should reconsider. This whole thing could blow up in our faces... (trails off, but is interrupted by the sound of footsteps or voices in the distance)"

```
conversation = ""
Person1: You're mine, you know. I made you.
Person2: I know.
Person1: If you went away, what would become of me?
Person2: I'm grown up now. I have to leave some time.
Person1: Of course you do, and I want you to... but there's no need to hurry it along, is there?
Person2: I can't help it.
Person1: That's twice this month you've slipped Deadly Night Shade into my tea and run off.
People might get the wrong idea and think you're unhappy at home.
""
```

Predicted Techniques: ['Persuasion or Seduction']

Chatbot Response[Given the detected technique of manipulation]:

" I understand your concerns, but I'm not trying to send any messages. I just need some space and time to figure out my own path. Let's talk about what's going on and find a way to communicate that works for both of us."

Chatbot Response:

" I'm not unhappy at home. I'm just... impatient. And I know you're not as fragile as you used to be, but I still want to make sure you're comfortable. Besides, it's not like I'm trying to hurt you. It's just a little... insurance policy. You know I'd never let anything happen to you."

```
conversation = ""  
Person1: You think I'm a fake.  
Person2: I think it's what you think.  
Person1: No, it isn't what I think.  
Person2: Look... You made a very calculated move, and then made me  
|feel embarrassed for responding to you. That wasn't necessary.  
""
```

Predicted Techniques: ['Playing Victim Role']

Chatbot Response[Given the detected technique of manipulation]:

"I think we're getting a bit sidetracked here. Let's focus on the actual issue at hand. Can we try to break down what happened and understand each other's perspectives without making assumptions about the other person's intentions?"

Chatbot Response:

Person2: Exactly. It felt like you were trying to provoke a reaction out of me, and then use it against me. That's not a very genuine way to interact with someone.

Declaration of use of AI

For this project we have used Gemini, ChatGPT and Claude.